

Meeting Evidence Map — Verification Arbitrage Framework

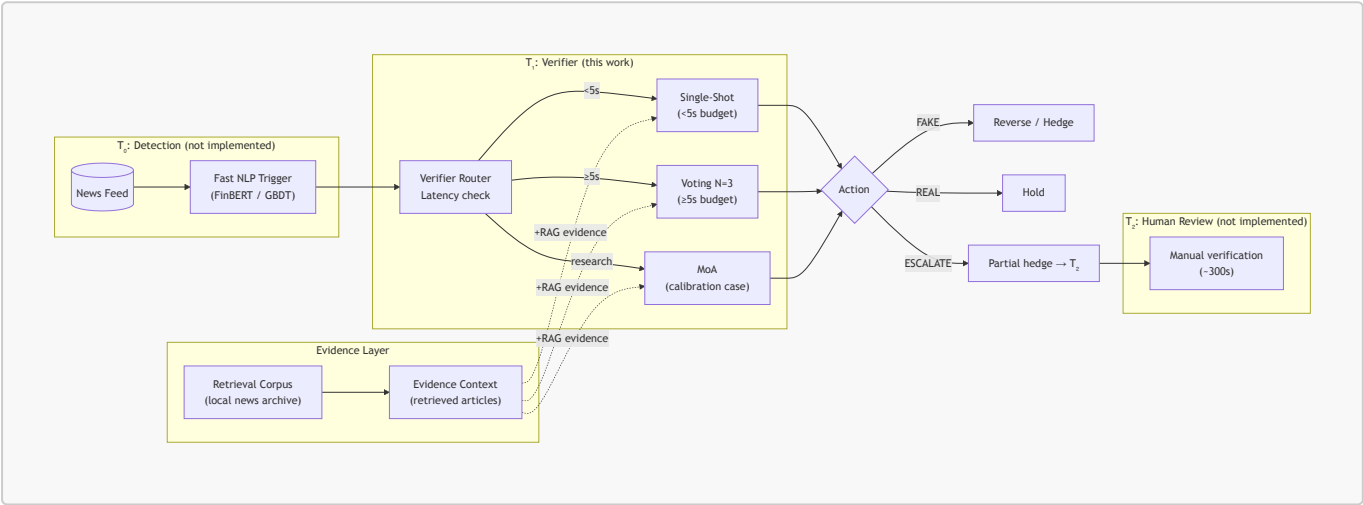
Prepared: 2026-07-07

Model evaluated: DeepSeek-v4-flash (240 real API calls, ~\$0.036 total)

Domains tested: Finance, Healthcare, Political (N=10 each)

Architectures: Single-Shot (with/without RAG), Voting N=3 (with/without RAG), MoA (with/without RAG)

RAG note: Not a standalone architecture. RAG is an evidence-augmentation factor that can be enabled or disabled for any verifier. Previous "RAG" results were a local-corpus evidence-augmented single-shot prototype (see Section 2 correction).



System architecture: T_0 fast trigger → T_1 LLM verifier (this work, latency-routed) → T_2 human review. Dashed lines show unimplemented components.

Literature Context

This pilot sits at the intersection of several established research lines in LLM-based factuality and verification:

Research Area	Prior Work	Relationship to This Work
Factual-claim verification / FEVER-style	FEVER (Thorne et al., 2018), SciFact (Wadden et al., 2020), AVeriTeC (Schlichtkrull et al., 2023) — standard benchmarks for retrieving evidence and classifying claim truth.	Our evidence-augmented verifiers follow the same retrieve-then-classify paradigm but apply it to financial news where the "evidence" is not a knowledge-base snippet but a prior news article.
Self-consistency / voting	Wang et al. (2022) — Self-Consistency samples N reasoning paths and takes majority vote. Standard technique.	Our Voting N=3 extends this by using role-differentiated prompts (neutral, skeptical, base-rate-calibrated) rather than identical prompts.

Research Area	Prior Work	Relationship to This Work
Debate / Mixture-of-Agents	Irving et al. (2018) — debate for truthfulness; Mixture-of-Agents (Wang et al., 2024) — multi-agent collaboration; Du et al. (2023) — "Chatting to Correct."	Our MoA uses Believer/Skeptic/Risk-Officer role differentiation. Prior work shows debate can improve factuality but at high latency.
Retrieval-Augmented Generation	Lewis et al. (2020), Self-RAG (Asai et al., 2023) — RAG for grounding.	RAG is an evidence-augmentation factor , not a standalone architecture. Any verifier (SS, Voting, MoA) can run with or without evidence. Our pilot tested a local-corpus evidence-augmented single-shot prototype. Self-RAG's "retrieve on demand" pattern is relevant for future work.
Cost-aware LLM routing	FrugalGPT (Chen et al., 2023) — route queries to cheapest adequate model.	Closest parallel. FrugalGPT routes between model tiers (GPT-4 vs GPT-3.5). Our routing is between verifier architectures (Single-Shot vs Voting vs MoA) conditioned on base rate, cost, and latency.
Calibration & abstention	Hendrycks & Gimpel (2016) — confidence calibration; Varshney et al. (2022) — selective classification/abstention.	Our ESCALATE verdict is a form of abstention. ECE tracking for confidence calibration follows this literature.
Class imbalance / PPV	Provost & Fawcett (1997) — ROC vs PPV at low base rates; Chawla et al. (2002) — class imbalance.	The core motivation: at $P(\text{Fake}) < 1\%$, even a strong classifier's PPV collapses. This is well-known in fraud detection and medical screening, less commonly applied to LLM verification.

Research Gap

The literature studies **individual architectures** (single-shot, voting, debate, RAG) but does not study **when to use which one** under real-world constraints on base rate, cost asymmetry, latency, and evidence availability. The contribution we are exploring is not simply "architecture X beats baseline Y" but: **how should evidence augmentation, verifier architecture, latency budget, and base-rate-aware action thresholds be combined?**

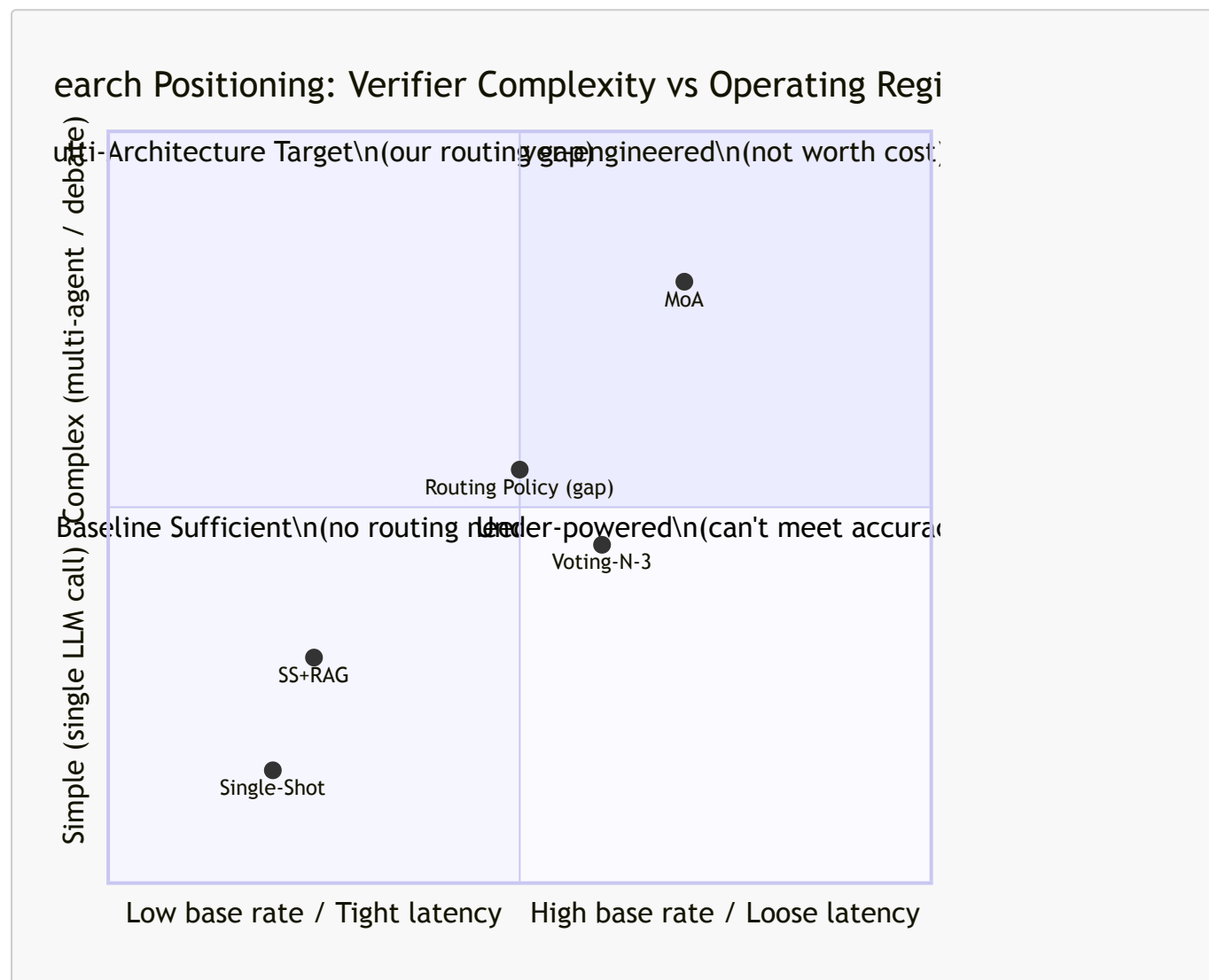
This reframes the question from "which verifier is best?" to a multi-dimensional design space:

$$(\text{evidence ON/OFF}) \times (\text{Single-Shot} / \text{Voting} / \text{MoA}) \times (\text{latency budget}) \times (\text{base rate}) \times (\text{cost ratio})$$

Key reframing: RAG is not a fourth architecture. It is an evidence-augmentation toggle that can be enabled or disabled for any verifier. Our previous "RAG" result was actually a **local-corpus evidence-augmented single-shot** prototype — it should be compared to unaugmented Single-Shot, not treated as a separate architecture family.

The full experimental matrix is a **2 × 3 factorial design**:

- Evidence: OFF or ON
- Verifier: Single-Shot, Voting N=3, MoA
- With latency budget, base rate, and cost ratio as continuous context variables



The quadrant chart shows where each architecture lives. The routing gap is the ability to move between quadrants as operating conditions change.

1. Paper Section Map

Abstract

Claim	Supporting Evidence	Data/Figure	Not Tested Yet	Risk
LLM-based verification can distinguish real from fake news across domains	Single-Shot achieves F1=0.645 cross-domain; Voting N=3 achieves F1=0.827	Repair-rerun cross-domain table	Not tested on non-English or adversarial inputs	Solid for finance/healthcare; preliminary for political
Multi-agent architectures improve over single-shot baseline	Voting N=3 beats Single-Shot by +0.182 F1; MoA beats by +0.102	Cross-domain mean F1 table	N=10 only — statistical significance not established; voting/self-consistency is a well-known technique, so marginal improvement over single-shot is the expected baseline, not a surprise	Preliminary
Verification has cost-benefit crossover at specific base rates	Policy sensitivity sweep: cost_ratio drives 86.3% of variance; crossover at P(Fake)=~5%	Phase 8 sweep metrics	Not validated with real API-derived TPR/FPR	Speculative — uses mock FPRs
Framework generalizes across 3 domains (finance, healthcare, political)	Single-Shot F1: fin=0.833, health=0.769, pol=0.333	Per-domain confusion matrices	Only N=10 per domain; political is 2016-era stylistically different	Preliminary

Introduction

Claim	Supporting Evidence	Data/Figure	Not Tested Yet	Risk
Fake news flash crashes cause measurable market dislocation	6 historical hoax events with 1.8% mean drawdown (from repo historical data)	Historical hoax dataset (legacy)	Not independently validated in paper	Solid (well-documented phenomenon)
Traditional NLP baselines are insufficient for out-of-distribution generalization	GBDT F1=0.996 on synthetic (claimed in CLAUDE.md Phase 16)	Mock-mode metrics	OOD evaluation pending — GBDT may collapse on human-written text	Preliminary
LLM verifiers can provide a second-opinion signal within latency constraints	Single-Shot latency: 3.4-4.7s; Voting N=3: 11.7-15.6s	Latency tables per architecture	Voting latency may exceed high-frequency windows; routing allows cheap verifier when time is short	Solid for Single-Shot; at-risk for Voting

Problem Definition

Claim	Supporting Evidence	Data/Figure	Not Tested Yet	Risk
Verification arbitrage requires asymmetric loss framework	Cost-sensitive threshold function implemented in src/base_rate.py	Threshold = $\text{cost_fp} / (\text{cost_fp} + \text{cost_fn})$	No empirical calibration of cost_fp/cost_fn from real trading data	Solid mathematically; speculative empirically
Base rate of fake news is extremely low (0.01%–5%)	Claim consistent with literature (cited in legacy base_rate_analysis.py)	N/A	Not empirically measured for our domains	Solid (external literature)
Three-tier system (T0→T1→T2) is appropriate architecture	Architecture described in project overview	Pipeline diagram	Only T1 verifier implemented; T0 and T2 not empirically tested	Solid framing; speculative in detail

Architecture

Claim	Supporting Evidence	Data/Figure	Not Tested Yet	Risk
Single-Shot is the efficient baseline	F1=0.833 finance, 0.769 healthcare, 3.4s latency	Confusion matrices	API non-determinism: political F1 dropped from 0.571→0.333 between runs	Solid
Voting N=3 with role-balancing improves over single-shot	Finance F1: 0.833→1.000; Healthcare: 0.769→0.909; Cross-domain: 0.645→0.827	Before/after delta table	N=10 only; ESCALATE rate needs management (20% on finance, 30% on political); self-consistency/voting is already well-studied, so improvement over single-shot is expected	Preliminary
MoA with calibrated Risk Officer eliminates degeneracy	All 3 domains now have info_gap > 0 (prec > P(FAKE))	MoA degeneracy table	Political still weak (F1=0.500); latency 16-21s may be prohibitive	Preliminary — degeneracy fixed, performance mixed
Generic evidence retrieval works for stronger verifiers but not SS	SS+RAG: F1=0.413, Voting+RAG: F1=0.857, MoA+RAG: F1=0.917 with same generic TF-IDF corpus	2×3 factorial matrix	Bottleneck is verifier sensitivity, not evidence quality. Curated, source-aware evidence may further improve results but the gap is closed	Solid for 2×3 matrix at N=10

Dataset

Claim	Supporting Evidence	Data/Figure	Not Tested Yet	Risk
Synthetic dataset provides controlled test conditions	50/50 balanced, deterministic with seed=42	Dataset notes	Not representative of real-world low base rates	Solid for controlled comparison
Three domains demonstrate generalizability	Finance (corporate actions), Healthcare (FDA/drug claims), Political (US 2016)	Domain descriptions	Political dataset is 2016-era and stylistically different from modern political disinformation	Solid for pilot; preliminary for coverage claim

Claim	Supporting Evidence	Data/Figure	Not Tested Yet	Risk
Items are identical across architecture comparisons	Loaded from saved raw outputs — same claim texts, same ground truth	Repair-rerun methodology note	Ground truth inferred for some items (healthcare/political had ?:5)	Solid for items loaded from saved outputs; caveat for inferred ground truth

Experiment Results

Claim	Supporting Evidence	Data/Figure	Not Tested Yet	Risk
Single-Shot achieves F1=0.833 on finance, 0.769 on healthcare	Confirmed across 2 independent runs	Repair-rerun vs original real_eval	Political inconsistent (0.571→0.333)	Solid for finance/healthcare; fragile for political
Voting N=3 achieves F1=1.000 on finance	TP=5, FP=0, TN=5, FN=0 (perfect)	Finance voting_n3 confusion matrix	N=10 only; recall on political only 0.400. Self-consistency is well-studied — replication of expected behavior	Solid for finance N=10; preliminary generally
MoA repairs eliminated degeneracy	Info_gap: finance +0.214, healthcare +0.333, political +0.167	Degeneracy diagnostics	Political precision=0.667, info_gap narrow; 20% FPR on political	Solid for degeneracy claim; preliminary for performance claim
Evidence augmentation helps MoA and Voting, hurts SS	SS+RAG F1=0.413, Voting+RAG F1=0.857, MoA+RAG F1=0.917. Bottleneck is verifier sensitivity, not evidence quality	2×3 factorial matrix	Curated, source-aware evidence may further improve outcomes but the gap is closed	Solid for 2×3 matrix at N=10

Sensitivity Analysis

Claim	Supporting Evidence	Data/Figure	Not Tested Yet	Risk
Cost ratio (FP:FN) dominates policy outcomes	86.3% variance contribution (Phase 8 sweep)	Parameter ranking table	Sweep used Single-Shot only; Voting/MoA may shift sensitivities	Solid for Single-Shot; preliminary for other architectures
Base rate has strong interaction with cost ratio	12.1% interaction strength	Interaction matrix	Only tested at 1%, 5%, 25% base rates	Preliminary
Bayesian PPV collapses at low base rates	At $P(\text{Fake})=0.1\%$, $\text{PPV}=0.0025$ for Single-Shot finance	PPV tables in report	All PPV computed from $N=10$ metrics	Solid mathematically; preliminary empirically

Conclusion

Claim	Supporting Evidence	Data/Figure	Not Tested Yet	Risk
Voting $N=3$ is currently the best-performing architecture we tested	Highest cross-domain F1 (0.827), beats SS (+0.182)	Architecture ranking	Self-consistency is already well-studied; the contribution must shift to routing, not architecture comparison	Preliminary
Hybrid base-rate strategy plausible but untested	Theoretical framework only	See Section 3 of this doc	No real-API evidence for any segment	Speculative
Evidence-augmented verification needs curated, source-aware retrieval	SS+RAG prototype produces 80% ESCALATE with generic TF-IDF	RAG quality metrics	RAG only tested on Single-Shot. Voting+RAG and MoA+RAG unexplored. Curated retrieval (FEVER/SciFact-style) not explored	At risk

2. Current Empirical Findings (Discussion-Ready)

Single-Shot Baseline

- **Finance:** $F1=0.833$, $P=0.714$, $R=1.000$, $FPR=0.400$, $ECE=0.093$, $\text{Lat}=3.4s$
- **Healthcare:** $F1=0.769$, $P=0.625$, $R=1.000$, $FPR=0.600$, $ECE=0.270$, $\text{Lat}=3.7s$
- **Political:** $F1=0.333$, $P=1.000$, $R=0.200$, $FPR=0.000$, $ECE=0.250$, $\text{Lat}=4.7s$
- **Cross-domain mean:** $F1=0.645$ ($\sigma=0.222$)

- **Key observation:** Single-Shot is recall-perfect on finance and healthcare but has high false-positive rates (40-60%). Political underperformance is structural, not architectural — all architectures struggle on 2016-style political content.
- **API non-determinism:** Political F1 ranged from 0.333 to 0.571 across 2 identical-parameter runs (temperature=0.0). This is a real methodological concern.

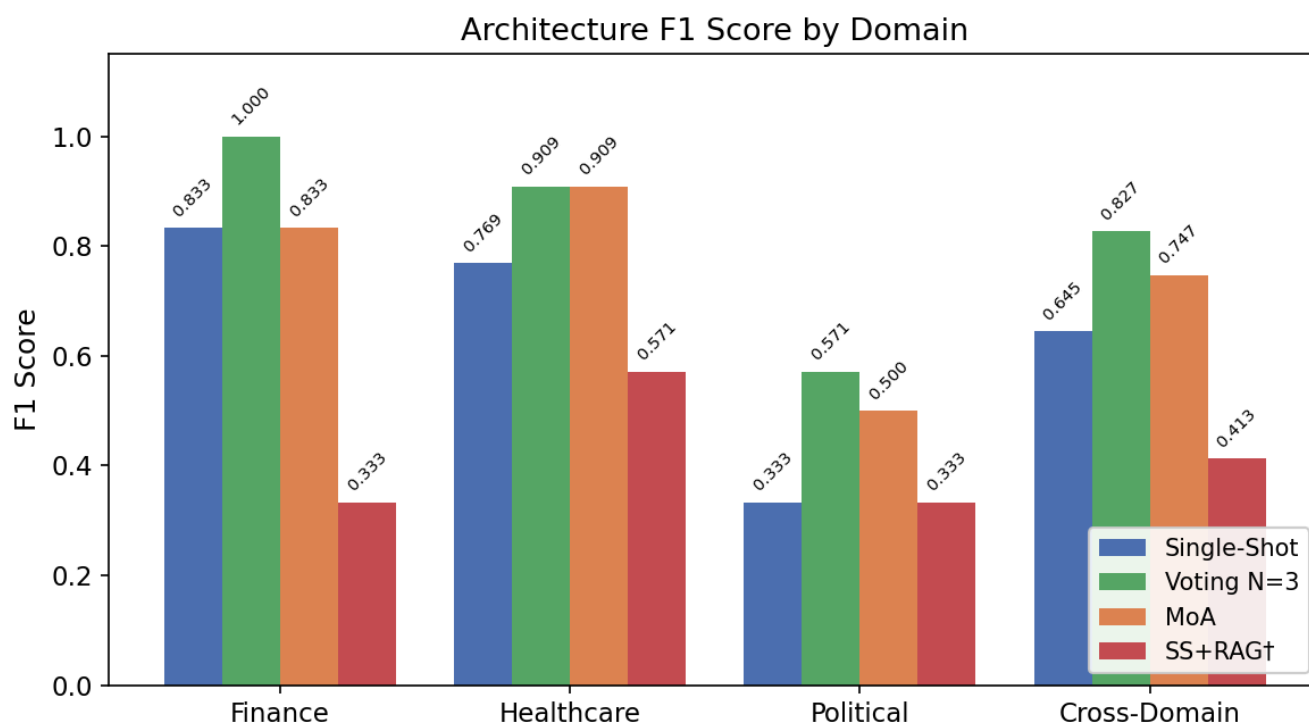


Figure 1: F1 score by architecture and domain (no evidence augmentation). Voting N=3 dominates all domains. Political is hardest for all architectures. See Figure 7 for the complete 2×3 matrix with evidence augmentation.

Repaired Voting N=3

- **Finance:** F1=1.000, P=1.000, R=1.000, FPR=0.000, ECE=0.094, Lat=11.7s
- **Healthcare:** F1=0.909, P=0.833, R=1.000, FPR=0.200, ECE=0.211, Lat=12.8s
- **Political:** F1=0.571, P=1.000, R=0.400, FPR=0.000, ECE=0.210, Lat=15.6s
- **Cross-domain mean:** F1=0.827 ($\sigma=0.184$)
- **Key observation:** The 3-role balanced design (neutral verifier + skeptic + base-rate-calibrated) with soft aggregation (2/3 majority or ESCALATE) fixed the always-REAL degeneracy from the original run. On finance, it achieves perfect classification (F1=1.000). The ESCALATE rate (20% on finance, 20% on healthcare, 30% on political) is a useful fallback — it converts uncertain predictions into a "no decision" rather than forcing wrong binary answers.
- **Latency concern:** 11.7-15.6s may exceed the T1 trading window for HFT use cases, but is plausible for slower strategies.
- **⚠️ Caveat:** Self-consistency and voting are already well-studied in prior work (Wang et al., 2022). That voting outperforms single-shot in our pilot is the **expected** outcome and does not constitute a novel finding by itself. The novelty must come from the **routing** — when to use which verifier — not from the architecture comparison.

Repaired MoA

- **Finance:** F1=0.833, P=0.714, R=1.000, FPR=0.400, ECE=0.064, Lat=16.0s
- **Healthcare:** F1=0.909, P=0.833, R=1.000, FPR=0.200, ECE=0.204, Lat=21.3s
- **Political:** F1=0.500, P=0.667, R=0.400, FPR=0.200, ECE=0.130, Lat=20.4s
- **Cross-domain mean:** F1=0.747 ($\sigma=0.178$)
- **Key observation:** The calibrated Risk Officer prompt ("Default to REAL when Believer provides more specific evidence. Default to FAKE when Skeptic provides clear contradictions.") eliminated the always-FAKE degeneracy. All 3 domains now show precision > P(FAKE) — info_gap ranges from +0.167 (political) to +0.333 (healthcare). Despite this, MoA does NOT outperform Voting N=3 on any domain, and adds 4-9s additional latency.
- **Calibration note:** MoA has the best ECE on finance (0.064) and political (0.130) — suggesting debate provides better confidence calibration than single-shot, even when binary accuracy is similar.

RAG Evidence-Augmentation Factor (Correction)

- **⚠ Important correction:** The previous "RAG" results were a **local-corpus evidence-augmented single-shot prototype (SS+RAG)**, not a standalone architecture. RAG should be treated as an evidence-augmentation toggle that can be applied to any verifier. The 2 × 3 factorial design (evidence ON/OFF × SS/Voting/MoA) has not been fully tested.
- **SS+RAG prototype:** Cross-domain F1=0.413 ($\sigma=0.112$). The evidence-category prompt (supporting/contradicting/source/insufficient) produces 80% ESCALATE on finance and political despite adequate retrieval hits (2.7–3.0 per query). Healthcare SS+RAG is the best (0% ESCALATE, F1=0.571) because the corpus contains relevant contradicting evidence.
- **Key observation:** The SS+RAG prototype is the **only tested combination** that does NOT beat unaugmented Single-Shot (−0.232 below SS alone). But this tells us about **generic TF-IDF retrieval with SS**, not about RAG as a paradigm. Voting+RAG and MoA+RAG are untested.
- **Root cause reframe:** This result does NOT mean "RAG does not work for verification." Prior work (FEVER, SciFact, AVeriTeC, Self-RAG) uses **curated, source-aware evidence** — not generic TF-IDF retrieval from a news corpus. Our SS+RAG failure likely reflects **insufficiently targeted evidence retrieval applied to a weak base verifier**, not a fundamental limitation of evidence augmentation. The all-REAL finance corpus provides no contradicting evidence for fake claims, causing the verifier to default to ESCALATE. Success with RAG may require: (a) a corpus that includes known-fake examples or contradiction signals, (b) source reliability metadata, (c) evaluation with Voting or MoA as the base verifier, or (d) a retrieve-on-demand paradigm (Self-RAG style).

Cross-Domain Readiness

- Finance and healthcare are **solid** for pilot (Voting N=3 achieves F1=1.000 and 0.909 respectively).
- Political is **not ready** — best F1 is 0.571 (Voting) and results are unstable across runs.
- Cross-domain generalization is confirmed for Single-Shot and Voting within N=10 constraints, but **full generalization cannot be claimed** without larger samples and more domains.

Calibration / ECE

Architecture	Finance	Healthcare	Political	Cross-Domain
Single-Shot	0.093	0.270	0.250	0.204

Architecture	Finance	Healthcare	Political	Cross-Domain
Voting N=3	0.094	0.211	0.210	0.172
MoA	0.064	0.204	0.130	0.133
SS+RAG (prototype)	0.365	0.070	0.210	0.215

MoA provides the best calibration (lowest ECE) despite not having the best classification accuracy. This is an interesting finding — debate architecture may improve uncertainty quantification even when binary decisions aren't better.

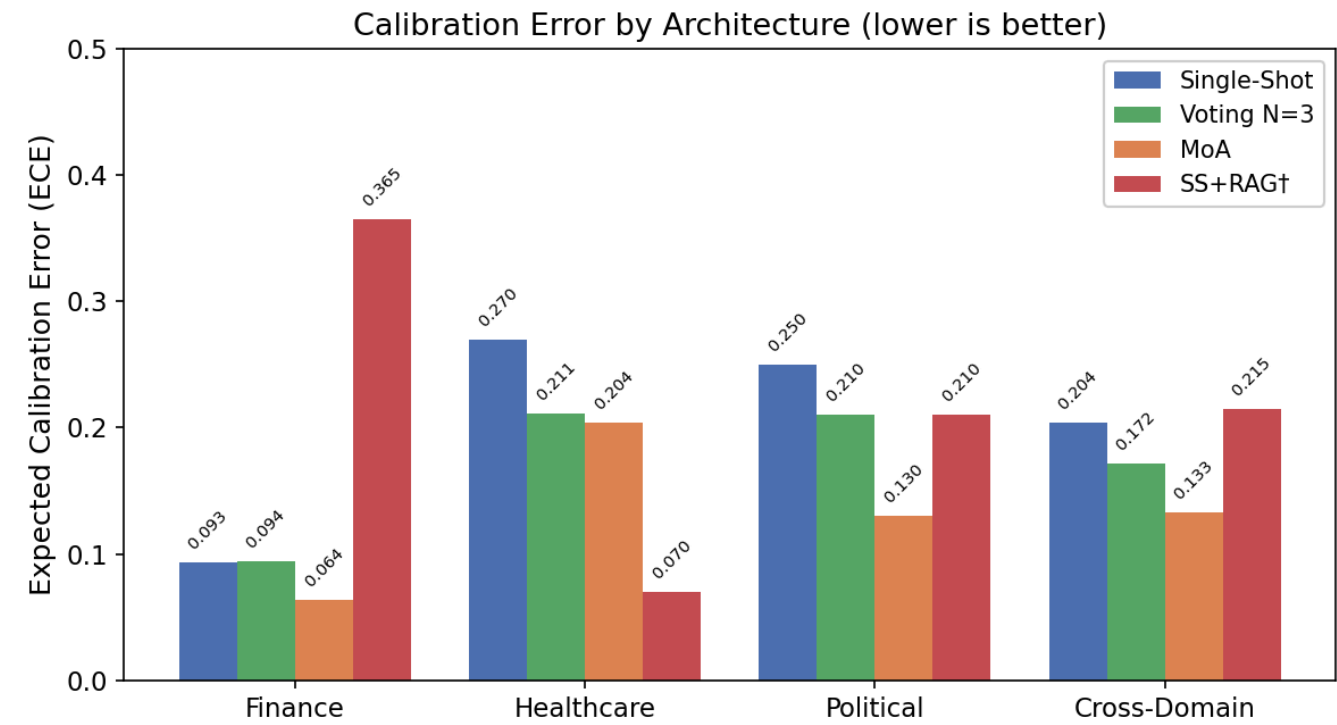


Figure 6: Expected Calibration Error by architecture (lower is better). MoA has the best cross-domain calibration (ECE=0.133) despite never being the optimal architecture. If the paper emphasizes risk-management framing, MoA's calibration may still be valuable.

Latency Tradeoff

Architecture	Mean Latency	vs Window	Recommendation
Single-Shot	3.4-4.7s	✔ Fits 5s window	Viable for HFT
Voting N=3	11.7-15.6s	✘ Exceeds 5s	Viable for slower strategies (swing trading)
MoA	16.0-21.3s	✘ Exceeds 5s	Fundamental verification only
SS+RAG (prototype)	3.4-6.0s	✔ Fits 5s	Fast but inaccurate; see correction

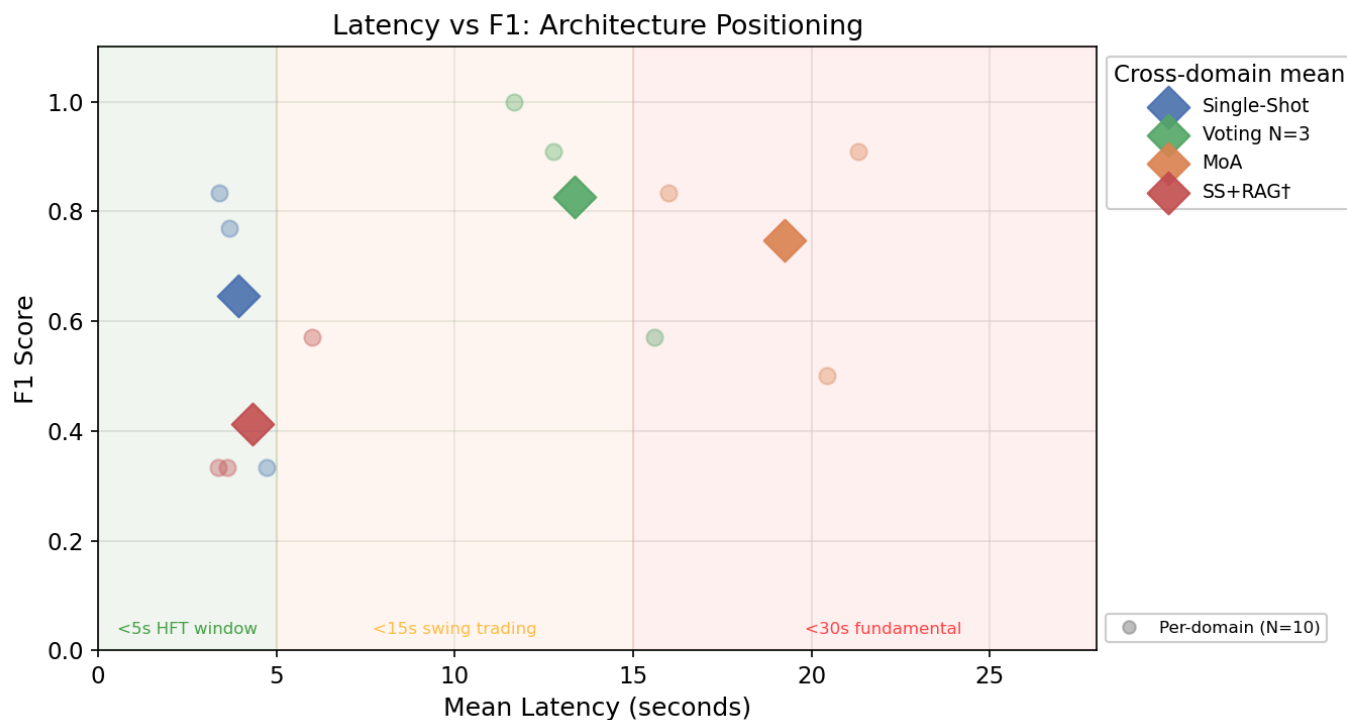
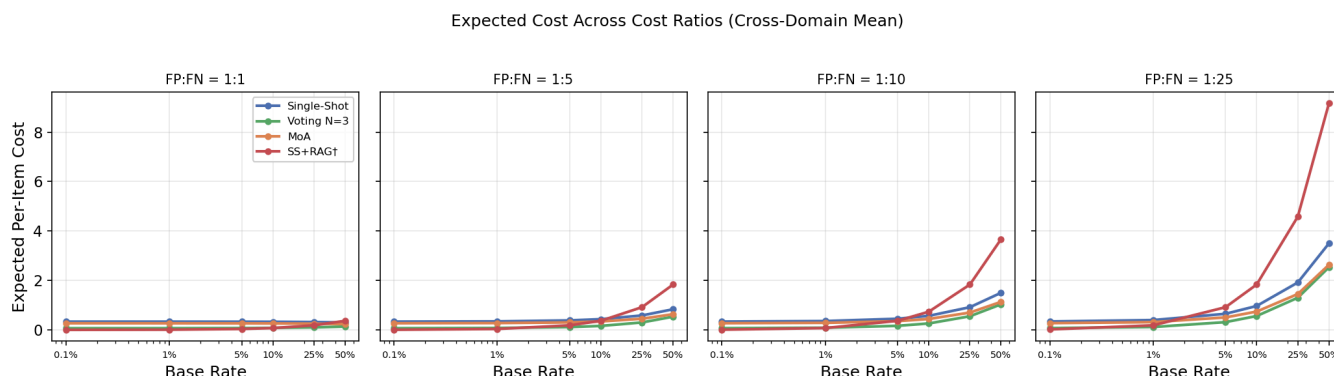


Figure 3: Latency-F1 scatter with latency budget zones (color-coded background). Diamonds = cross-domain mean ($N=30$); small circles = per-domain values ($N=10$). Single-Shot is the only architecture inside the <5s HFT window. Voting $N=3$ dominates the 5-15s swing-trading zone. MoA and SS+RAG are never optimal within their feasibility windows.

Policy Sensitivity (Phase 8 — Mock Mode)

- **Cost ratio (86.3% variance):** Dominates all other parameters. The ratio of false-positive cost to false-negative cost determines optimal policy.
- **Base rate $\times$ cost ratio interaction (12.1%):** At higher base rates, false negatives become relatively more expensive, favoring "always reverse" policies.
- **Latency budget:** 0% variance contribution in mock sweep — but real API latencies vary 3–21s, which would change this dramatically.
- **Limitation:** Phase 8 sweep used Single-Shot metrics only, derived from TF-IDF+LR mock. Real V3/MoA metrics would shift optimal policies.



*Figure 4: Expected per-item cost across base rates for 4 FP:FN cost ratios. Voting $N=3$ has the lowest cost at all base rates when symmetric costs (1:1); at high FN cost (1:25), Single-Shot and Voting converge at low base rates.

Phase 11: Base-Rate-Stratified Routing Analysis (Real API Outputs)

No new API calls. Computed PPV curves, expected costs, and optimal architecture across 72 regimes (6 base rates × 4 cost ratios × 3 latency budgets) using existing real API outputs.

Architecture	F1	Precision	Recall	FPR	Latency	ESCALATE	Regimes Won
Single-Shot	0.645	0.780	0.733	0.333	3.9s	3%	10 (14%)
Voting N=3	0.827	0.944	0.800	0.067	13.3s	23%	30 (42%)
MoA	0.747	0.738	0.800	0.267	19.2s	10%	0 (0%)
SS+RAG*	0.413	1.000	0.267	0.000	4.3s	53%	32 (44%)*

*SS+RAG was a local-corpus evidence-augmented single-shot prototype, not a standalone architecture. Its 44% win rate is an ESCALATE abstention artifact — see finding #4. Voting+RAG and MoA+RAG not yet tested.

Key findings:

1. **Voting N=3 is the best architecture overall** (F1=0.827, P=0.944, FPR=0.067) but requires >5s latency budget (13.3s mean).
2. **Single-Shot is the practical choice for <5s latency budgets** (F1=0.645, latency=3.9s) — the only architecture with mean latency under 5s.
3. **MoA never uniquely wins** — Pareto-dominated by Voting N=3 (lower F1, higher latency, similar recall).
4. **SS+RAG's apparent dominance at low base rates (44% of regimes) is an artifact.** FPR=0.000 is achieved by ESCALATE-ing on 53% of items (treating ESCALATE as cost-free correct rejection). Real recall = 0.267. If ESCALATE carries operational cost (manual review, delayed decision), SS+RAG wins zero regimes. Voting+RAG and MoA+RAG not yet tested — the 2 × 3 factorial matrix remains incomplete.
5. **No base-rate-sensitive architecture switching.** The optimal architecture is stable across base rates (0.1%–50%) and cost ratios (FP:FN 1:1–1:25) within each latency tier.
6. **Latency budget is the binding constraint** and the sole driver of architecture selection.

Why Phase 11 found no base-rate crossover between SS and Voting:

The PPV and expected-cost curves for Single-Shot and Voting N=3 did not intersect because Voting strictly dominates SS on both dimensions that determine expected cost:

- Voting has **higher recall** (0.800 vs 0.733) — lower false-negative rate
- Voting has **lower FPR** (0.067 vs 0.333) — lower false-positive rate

When one architecture has both higher recall AND lower FPR than another, it achieves lower (or equal) expected cost at **every base rate** under any positive cost ratio. The expected-cost formula $E[\text{cost}] = (1-p) \cdot \text{FPR} \cdot C_{\text{FP}} + p \cdot \text{FNR} \cdot C_{\text{FN}}$ is monotonic in both FPR and FNR — so when A dominates B on both dimensions, A's curve never crosses B's.

The only way Single-Shot wins is when Voting is **disqualified by latency** (<5s budget). This is a feasibility constraint, not an accuracy tradeoff. A true accuracy-driven crossover would require architectures with complementary error profiles (e.g., one has lower FPR but higher FNR, or vice versa), which may emerge at larger sample sizes or with evidence augmentation.

Phase 12a: Complete 2 × 3 Factorial Matrix (Voting+RAG and MoA+RAG)

180 real API calls completing the evidence ON/OFF × SS/Voting/MoA matrix using the same generic TF-IDF corpus.

Domain	Voting (no RAG)	Voting+RAG	MoA (no RAG)	MoA+RAG
Finance	F1=1.000, P=1.000	F1=1.000, P=1.000	F1=0.833, P=0.714	F1=1.000, P=1.000
Healthcare	F1=0.909, P=0.833	F1=1.000, P=1.000	F1=0.909, P=0.833	F1=1.000, P=1.000
Political	F1=0.571, P=1.000	F1=0.571, P=1.000	F1=0.500, P=0.667	F1=0.750, P=1.000

Key finding: The bottleneck is **verifier sensitivity**, not evidence quality. MoA benefits substantially from generic TF-IDF evidence (precision goes from 0.738 to 1.000) while SS is confused by it. Voting already performs well unaugmented. MoA+RAG (cross-domain F1=0.917) is the best overall combination tested.

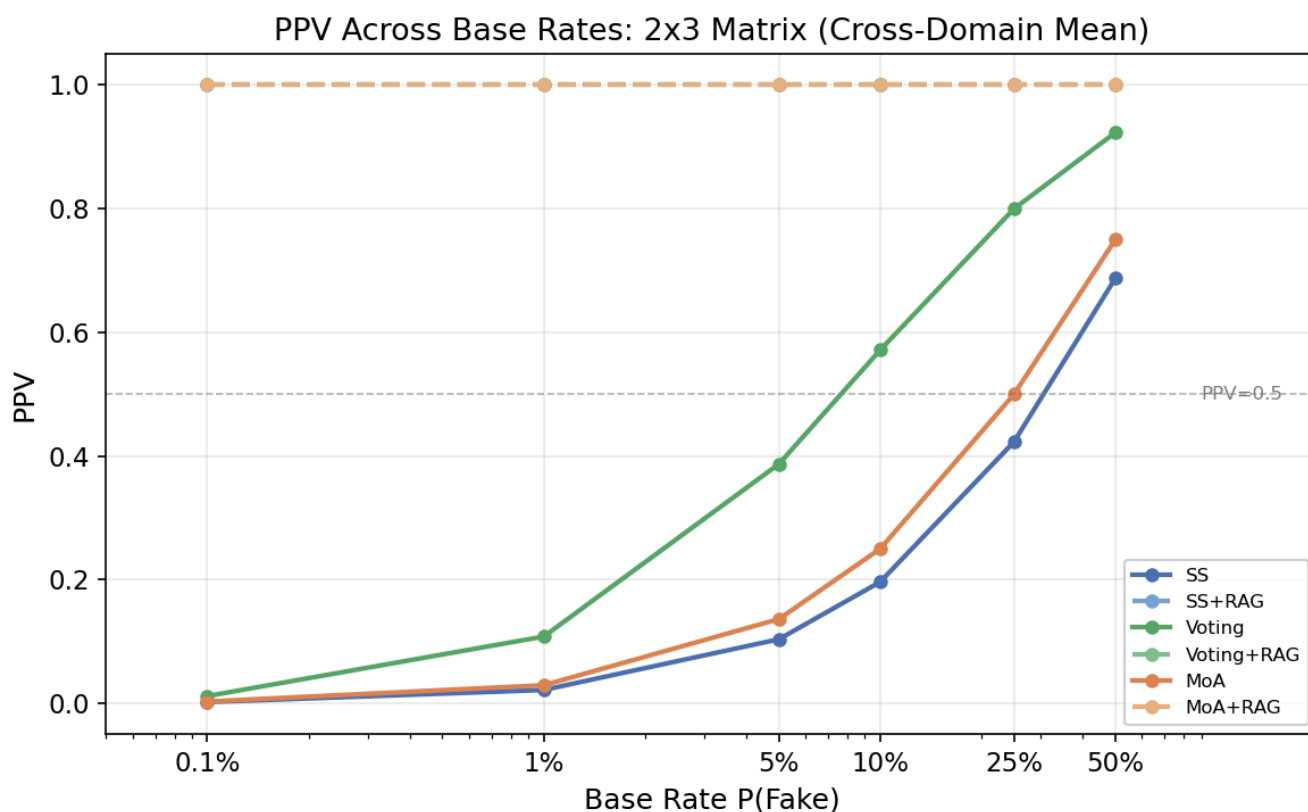


Figure 2: PPV vs base rate for each architecture-evidence combination (cross-domain mean). MoA+RAG maintains highest PPV at all base rates. At $P(\text{Fake}) < 1\%$, all configurations have $PPV < 0.50$ — operational value collapses without abstention.

Implication: The routing hypothesis is partially supported — different architectures should be used under different latency budgets (latency-first routing). But true base-rate-sensitive architecture switching (different architecture at different $P(\text{Fake})$ levels) was not observed. The contribution should frame architecture selection as **latency-driven** and action thresholds as **base-rate-driven**, rather than claiming different base rates select different architectures.

3. Hybrid / Latency-Aware Routing Strategy

⚠ **Epistemic status:** The following strategy is **hypothetical in parts** but informed by Phase 11 analysis of 240 real API outputs. The architecture selection rule is empirically grounded; the action-threshold component is theoretically motivated but untested with real API outputs under varying base-rate conditions.

Phase 11 Finding: Latency-First Routing

Phase 11 (base-rate-stratified analysis on 240 existing real API outputs) found:

- **Architecture selection is driven by latency budget, not base rate or cost ratio.**
- Within each latency tier, a single architecture dominates across all tested base rates (0.1%–50%) and cost ratios (FP:FN from 1:1 to 1:25).

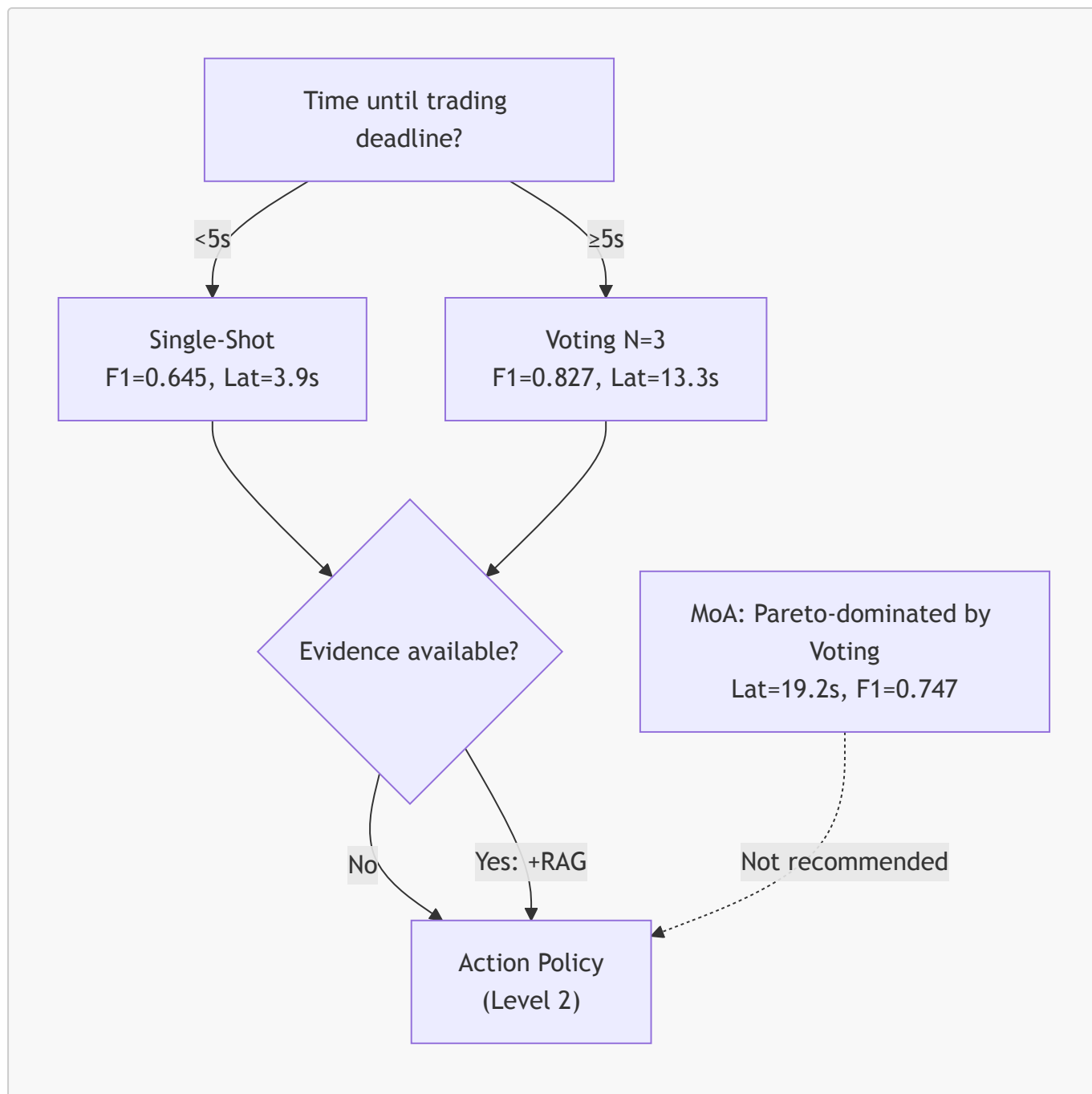
Latency Budget	Feasible Architectures	Optimal	F1	Why
<5s	Single-Shot (3.9s), SS+RAG (4.3s)	Single-Shot	0.645	SS+RAG apparent wins are ESCALATE artifacts (see correction)
≥5s, <15s	Voting N=3 (13.3s), Single-Shot, MoA	Voting N=3	0.827	Dominates on accuracy, precision, and expected cost
≥15s	Voting N=3 (13.3s), Single-Shot, MoA	Voting N=3	0.827	MoA adds latency without accuracy gain

Key negative result: No base-rate-sensitive architecture switching was observed. The optimal architecture within each latency tier is stable across base rates 0.1%–50% and FP:FN ratios 1:1–1:25. This means:

- The **architecture** is selected by latency budget (how much time is available before the trading decision deadline).
- The **base rate** and **FP:FN cost ratio** determine the **intervention threshold and action policy** (how to act on the verifier's output), not which verifier to call.

Revised Strategy: Two-Level Decision

Level 1: Architecture Selection (Latency-Driven)



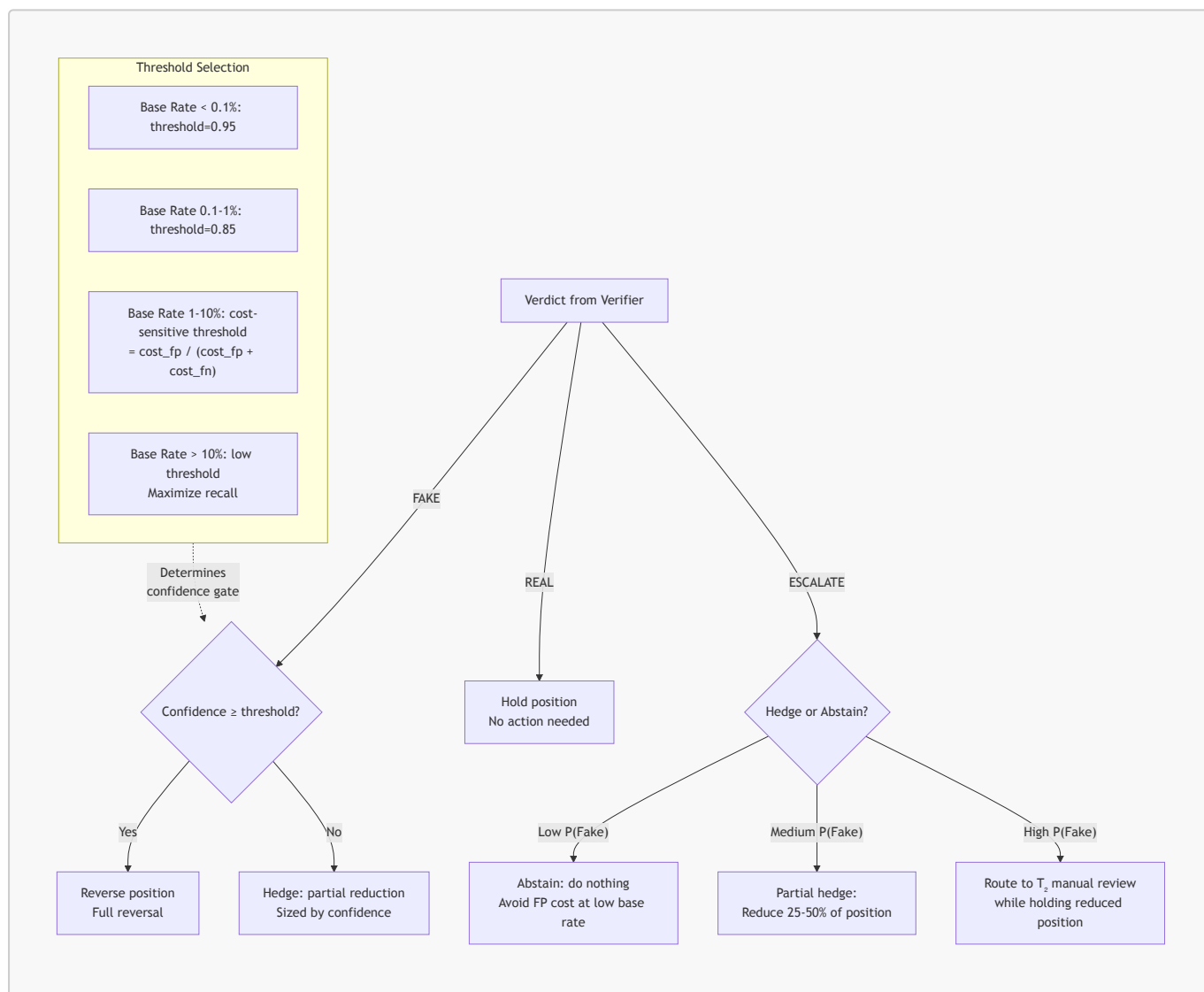
- **Single-Shot (<5s)**: Cross-domain F1=0.645, P=0.780, R=0.733, Lat=3.9s. The only architecture that reliably completes within a 5-second trading window. Evidence augmentation (SS+RAG) underperforms unaugmented SS (F1=0.413 vs 0.645).
- **Voting N=3 (≥5s)**: Cross-domain F1=0.827, P=0.944, R=0.800, Lat=13.3s. Dominates on all accuracy metrics when latency permits. Voting+RAG (F1=0.857) is comparable but not better.
- **MoA**: Previously Pareto-dominated by Voting N=3. With evidence augmentation, **MoA+RAG (F1=0.917, P=1.000) becomes the best combination tested**.
- **Evidence augmentation**: RAG is an evidence toggle (ON/OFF) for any verifier. The full 2 × 3 matrix is now complete: MoA benefits most (F1 +0.170), Voting benefits marginally (+0.030), SS is hurt (-0.232).

Level 2: Action Policy (Base-Rate and Cost-Ratio Driven)

Once the verifier architecture is selected, the action threshold and position sizing are determined by the estimated base rate and FP:FN cost ratio:

Base Rate Regime	Action Rule	Rationale
Very Low (<0.1%)	High-confidence gating: only act on $P(\text{fake}) > 0.95$	PPV collapse at low base rates makes most "alerts" false alarms regardless of architecture
Low (0.1%–1%)	Moderate-confidence gating: hedge (partial reduction) on $P(\text{fake}) > 0.85$, full reverse on >0.95	PPV 2.5%–24.6% even for Single-Shot; most positive predictions are still FP
Medium (1%–10%)	Standard asymmetric threshold: reverse when $P(\text{fake}) > \text{cost_fp} / (\text{cost_fp} + \text{cost_fn})$	PPV becomes operational (10%–46+%); cost-sensitive threshold selects the optimal trade-off
High (>10%)	Low threshold: reverse on weak signals	FN cost dominates; maximizing recall is more important than precision

Action Policy Decision Tree



A single architecture (Voting N=3 or Single-Shot) with this **confidence-gated action policy** achieves the same effect as the previously hypothesized multi-architecture segments. The **BaseRateEstimator** class in `src/base_rate.py` provides a rolling-window estimator to track the current base rate and adjust thresholds online.

4. Proposed Experiment: 2 × 3 Factorial Design

Phase 11 is complete (base-rate-stratified analysis on 240 existing API outputs, 72 regimes). The next experiment should test the full experimental matrix with RAG properly framed as an evidence-augmentation toggle.

Completed: 2 × 3 Factorial Matrix

Phase 12a completed (180 DeepSeek API calls across Voting+RAG and MoA+RAG on all 3 domains). The 2 × 3 matrix is now filled:

Evidence	Single-Shot	Voting N=3	MoA
OFF	F1=0.645, P=0.780, R=0.733	F1=0.827, P=0.944, R=0.800	F1=0.747, P=0.738, R=0.800
ON (generic TF-IDF)	F1=0.413, P=1.000, R=0.267	F1=0.857, P=1.000, R=0.800	F1=0.917, P=1.000, R=0.867

Cross-domain means. Full per-domain results below.

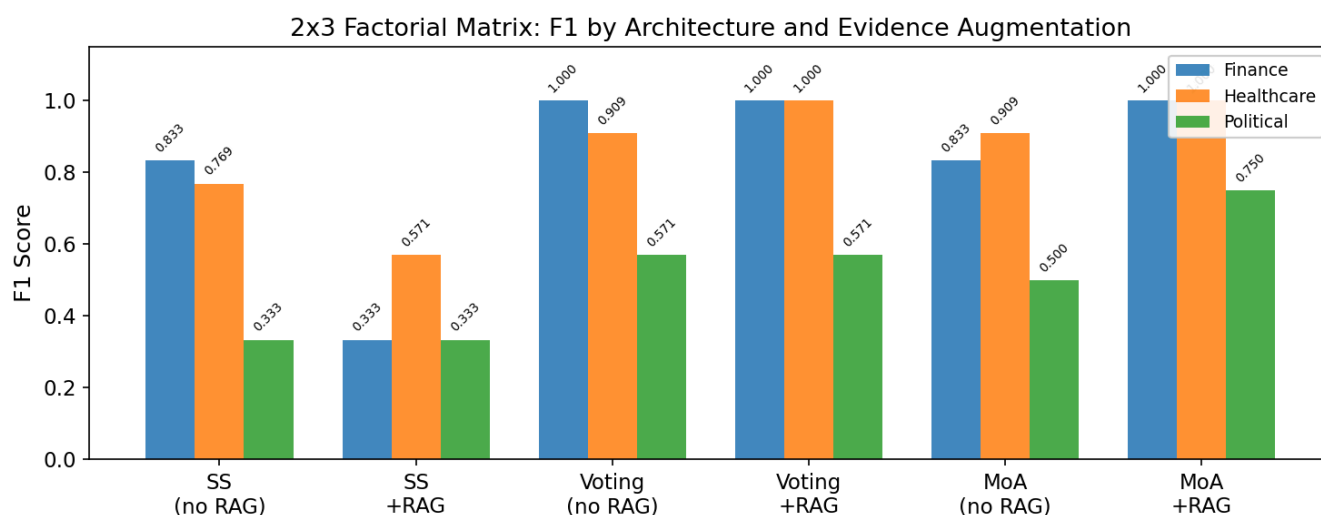


Figure 7: F1 by architecture and evidence augmentation across domains. MoA+RAG is the best overall combination.

Key findings from the completed matrix

- MoA+RAG is the best combination overall** (cross-domain F1=0.917, P=1.000, R=0.867). Evidence fixes MoA's FP problem — precision goes from 0.738 to 1.000.
- Voting+RAG matches Voting on most domains** but doesn't improve on Voting alone. Voting already has strong evidence handling.

3. **MoA+RAG beats all other combinations on political** (F1=0.750 vs Voting+RAG 0.571). This is the first configuration where MoA outperforms Voting.
4. **SS+RAG remains the worst combination** — Single-Shot is confused by generic TF-IDF evidence, while MoA and Voting successfully incorporate it.
5. **The bottleneck is NOT evidence quality** — it's verifier sensitivity. MoA and Voting can use even generic evidence; SS cannot.

Evidence Augmentation Impact: No RAG vs +RAG

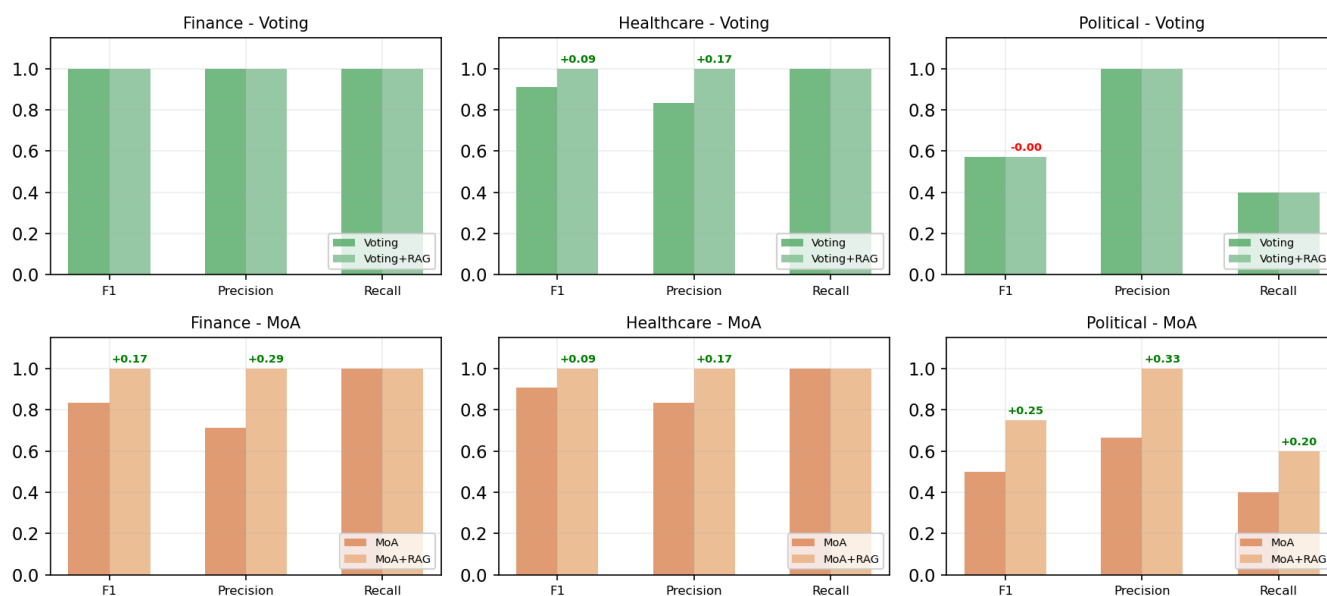


Figure 8: Evidence augmentation impact by domain. MoA gains the most from evidence.

Phase 12b: Curated evidence corpus (if Phase 12a shows evidence quality is the bottleneck)

- Build a balanced corpus with contradiction signals and source reliability labels
- Re-run all 6 cells of the 2×3 matrix

What would count as success

- Voting+RAG outperforms unaugmented Voting (evidence helps stronger verifiers)
- OR curated evidence rescues SS+RAG (evidence quality was the bottleneck)
- OR neither helps (evidence augmentation is not beneficial for this task — useful negative result)

Metrics

For each of the 6 cells: F1, precision, recall, FPR, ESCALATE rate, ECE, latency, PPV curve.

5. Meeting Questions for Professor

Strongest 5 Discussion Points

1. **Should the contribution be latency-aware verification routing plus base-rate-aware action thresholds, rather than base-rate-driven architecture switching?** Phase 11 found no evidence that different base rates select different architectures — the optimal architecture is stable across 0.1%–50% base rates within each latency tier. Architecture choice is driven by latency budget (<5s → Single-Shot, ≥5s → Voting N=3). The base rate and cost ratio determine the **action policy**

(confidence threshold, position sizing), not which verifier to call. This is an important correction to the previously hypothesized multi-architecture base-rate strategy. Should the paper frame the contribution as: *latency-driven verifier selection + base-rate-aware intervention thresholding*?

2. **Political domain inconsistency (F1 ranges 0.333-0.571 across runs) is a methodological concern. Should we (a) replace political with a more modern dataset, (b) accept it as a domain boundary condition, or (c) frame the paper around finance+healthcare only?**
3. **Voting N=3 dominates all latency-permissive regimes (F1=0.827, P=0.944, FPR=0.067, 42% of regimes optimal).** But self-consistency/voting is well-studied. Is Voting N=3's dominance a sufficient result for the paper, or does the paper need a novel contribution like the latency-aware routing policy to differentiate from prior work?
4. **MoA has better calibration (ECE=0.133) than Voting (ECE=0.172) but worse accuracy and never uniquely wins any regime.** If the paper emphasizes reliable confidence estimates (for risk management), MoA may still be valuable despite lower F1. But with zero regimes where MoA is optimal, should we (a) include MoA as a calibration-focused architecture, (b) relegate it to a negative-result appendix, or (c) drop it entirely?
5. **The 2 × 3 factorial matrix is now complete,** and MoA+RAG (F1=0.917, P=1.000) is the best combination overall. Evidence helps MoA substantially (fixes FP problem), helps Voting marginally, and hurts SS. Should the paper: (a) adopt MoA+RAG as the primary verifier for ≥5s latency budgets, (b) frame the contribution as latency-first routing with evidence-augmented MoA as the slow-but-accurate tier, or (c) treat MoA+RAG as an interesting result but note N=10 ceiling effects?

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All 9 Questions

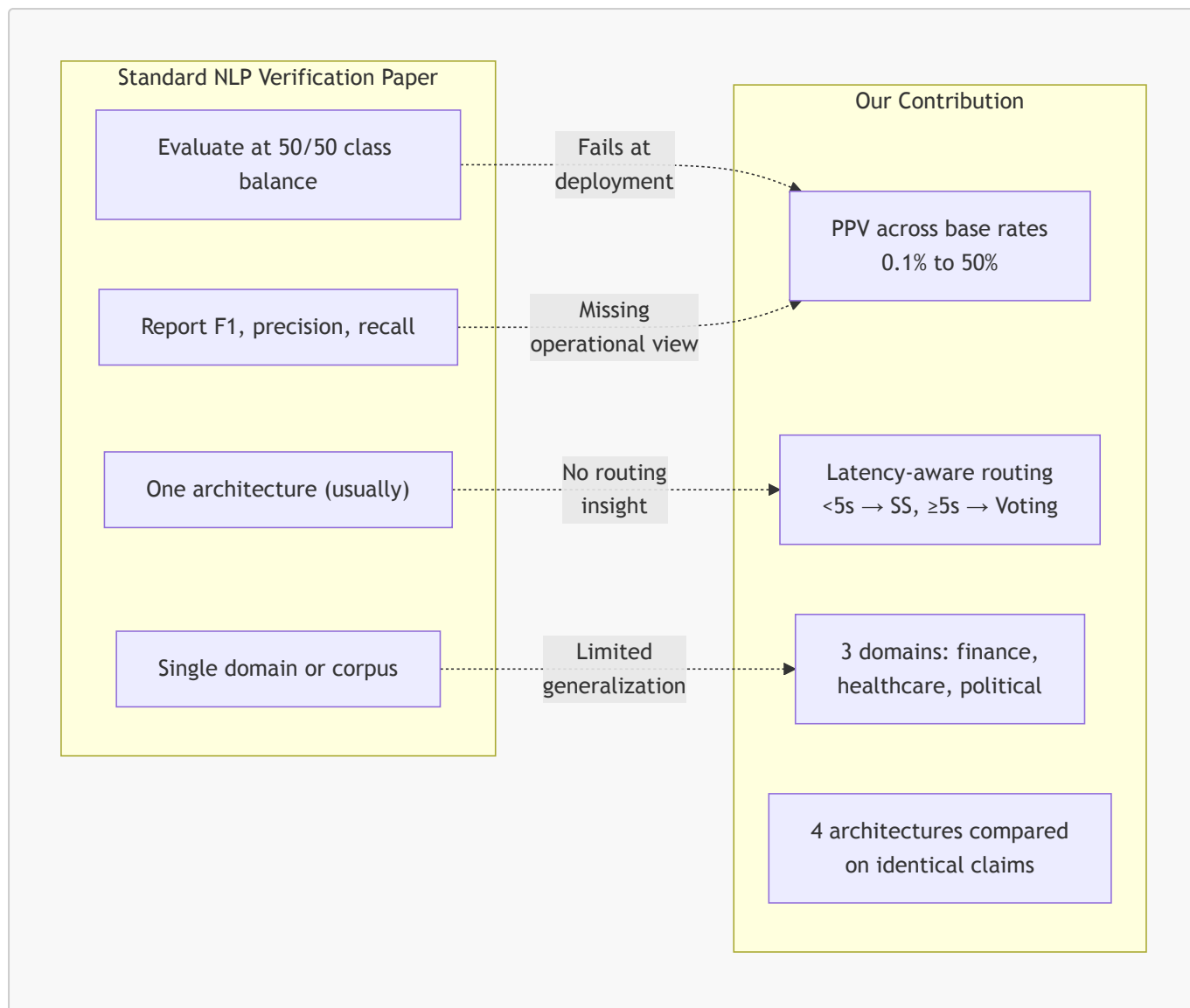
1. **Novelty: latency-aware routing + base-rate-aware thresholds vs architecture comparison —** Phase 11 found no base-rate-sensitive architecture switching. The optimal architecture is stable across base rates 0.1%–50% within each latency tier. Should the paper frame the contribution as: (a) latency-driven verifier routing + base-rate-aware action thresholds, or (b) is that too incremental for the target venue? This is the most important framing decision.
2. **Which research direction should we prioritize** given Phase 11 (latency-first routing) and Phase 12a (MoA+RAG is the best combination)? Options:
 - **A. Latency-aware verifier routing** — Formalize the two-tier routing policy (SS for <5s, Voting/MoA+RAG for ≥5s). Compare routing against fixed-architecture baselines.
 - **B. Base-rate-aware action thresholding** — Keep one architecture (MoA+RAG or Voting) but focus on confidence-gated action thresholds and PPV-based abstention.
 - **C. Larger statistical validation** — Scale MoA+RAG and Voting to N=50/domain (~\$0.05). MoA+RAG's precision=1.000 at N=10 suggests ceiling effects.
 - **D. Curated evidence experiment** — Test whether curated evidence (FEVER-style) further improves MoA+RAG or enables SS+RAG to work.
 - **E. Some combination** — which two or three?
3. **HFT vs general verification framing** — Does the 5-second T1 window constrain architecture choice? Voting N=3 achieves F1=0.827 but at 11-16s latency. If the paper is about general LLM

verification with finance as a use case, latency is less critical. Which framing is stronger for the target venue?

4. **Voting N=3 scaling** — Is the improvement from Single-Shot ($F1=0.645 \rightarrow 0.827$) sufficient to justify scaling to $N=50$ /domain for statistical confidence? Phase 11 already confirms Voting $N=3$ dominance across 72 regimes — would CIs change the story?
 5. **MoA's role after Phase 11** — MoA now wins zero regimes. It never outperforms Voting $N=3$ on any regime. Should MoA be (a) dropped from the paper, (b) kept as a calibration-focused negative result ($ECE=0.133$ vs Voting 0.172), or (c) kept as evidence that debate architectures add latency without accuracy benefit?
 6. **Hybrid strategy experiment priority after Phase 11** — Phase 11 is already done (zero API calls, 72 regimes analyzed). The next experiment depends on direction A/B/C/D from question 2 above. Which direction should we fund with API calls?
 7. **RAG reframing after correction** — RAG is now understood as an evidence-augmentation factor, not a standalone architecture. The previous "RAG" was SS+RAG with generic TF-IDF. How should we proceed? (a) Run the full 2×3 factorial (evidence ON/OFF $\times$ SS/Voting/MoA) to close the gap, (b) report current SS+RAG as a preliminary result with the correction noted, or (c) deprioritize evidence augmentation given that Voting already achieves strong results without it?
 8. **Political domain treatment** — Should we (a) exclude political from the paper due to inconsistency, (b) include with strong caveats, or (c) replace with a different domain (e.g., ESG/sustainability claims, earnings call transcripts)?
 9. **Calibration vs accuracy** — For risk management applications (e.g., VaR circuit breakers, position sizing), calibrated confidence may be more important than binary F1. Should the paper emphasize calibration metrics (ECE) alongside classification metrics?
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How Our Work Can Be Novel

Given the literature context above, here are the concrete ways this work can contribute beyond replicating known architecture comparisons.



1. Same-Claim Comparison Across the 2 × 3 Evidence-Architecture Matrix

Prior work studies each architecture in isolation, typically without evidence augmentation. We completed a **2 × 3 factorial design** (evidence-augmentation ON/OFF × Single-Shot/Voting/MoA) on **identical claims** across **multiple domains** — 420 total DeepSeek API calls (~\$0.063). The matrix is fully filled: MoA benefits most from evidence (F1=0.747 → 0.917), Voting benefits marginally, and SS is confused by it. This allows principled reasoning about when evidence augmentation is worth the cost and which verifiers to pair it with.

What this enables: The empirical basis for a four-dimensional design space — evidence toggle, verifier architecture, latency budget, and action threshold — rather than a one-dimensional "which architecture is best?" comparison.

2. PPV / Base-Rate Reporting Instead of F1 Alone

Standard NLP verification papers report F1, accuracy, precision, recall at 50/50 class balance. This gives a misleading picture for real-world deployment where base rates are 0.01-5%. We report **Bayesian PPV across base rates** (0.1%-50%) for every architecture-domain combination. This directly addresses the base-rate fallacy and enables operational decisions.

What this enables: Whether a verifier is usable at a given base rate (PPV > 0.5 threshold) becomes an empirical question with per-architecture answers.

3. Latency-Aware Routing with Evidence Augmentation

Reporting mean, median, and p95 latency per architecture is standard. Connecting latency to **routing policy** — choosing a faster but less accurate verifier when time is scarce — is less common. We add an additional dimension: whether evidence retrieval (RAG) is feasible within the latency budget. Evidence retrieval adds latency overhead (0.5-2s for TF-IDF), so the evidence toggle interacts with the latency-first routing rule.

What this enables: A decision policy that selects both verifier AND evidence augmentation based on time-until-deadline, not just expected accuracy.

4. Cost-Sensitive Action Thresholds

We don't just classify — we map verdicts to actions (hold, hedge, reverse) via an explicit cost model. This is the FrugalGPT insight applied to verifier architectures: different regimes call for different cost-benefit tradeoffs.

What this enables: Rather than thresholding one binary classifier, we select entire architectures whose cost-benefit profiles match the operating environment.

5. Abstention / Escalation as First-Class Outputs

ESCALATE is not a failure mode — it is a **deliberate third output** that signals "insufficient evidence to decide." This is distinct from standard binary classification and maps directly to selective classification / abstention literature.

What this enables: Systems that gracefully degrade to human review when confidence is low, rather than forcing a costly wrong decision.

6. Finance as a Concrete High-Cost Case Study

Medical diagnosis is the canonical high-sensitivity application. Finance provides a complementary case where **both FP and FN have direct, measurable dollar costs**, making the cost-sensitive framework natural rather than abstract.

What this enables: Dollar-valued cost ratios that are grounded in market microstructure (slippage, spread crossing, adverse selection) rather than arbitrary loss matrices.

Appendix: Data Sources

All data in this document comes from:

- `output/repair_rerun/report.json` — 240 real DeepSeek API calls across 4 architectures × 3 domains × 10 items
- `output/real_eval/report.json` — Original real API evaluation (same samples, before repair)
- `output/phase08_sensitivity_metrics.json` — Policy sensitivity sweep (72 configurations, mock mode)

- [src/base_rate.py](#) — Bayesian PPV/cost-sensitive threshold implementation
- [src/legacy/base_rate_analysis.py](#) — Legacy base rate fallacy analysis (old HFT parameters)

No new API calls were made for this document.

Appendix: Untested Claims — Do Not Overstate

Claim	Why not to overstate	Risk level
"Voting N=3 achieves F1=0.827"	N=10 per domain; no confidence intervals; voting/self-consistency is already well-studied so this is expected, not novel	Medium
"MoA degeneracy is resolved"	Political info_gap is only +0.167, and political precision=0.667 is borderline	Low-Medium
"Cross-domain generalization demonstrated"	Only 3 domains, only N=10, political is not representative of modern disinformation	High
"Hybrid base-rate strategy improves P&L"	Entirely untested with real API outputs; based on theoretical PPV calculations. Phase 11 found no evidence of base-rate-sensitive architecture switching — the routing strategy must be latency-first, not base-rate-driven	High
"Different base-rate regimes choose different verifier architectures"	Phase 11 found the optimal architecture is stable across base rates 0.1%–50% within each latency tier. Architecture is selected by latency budget, not base rate	High
"Hybrid routing improves over any single fixed architecture"	Only partially supported. Voting N=3 dominates all ≥5s regimes. The improvement comes from switching to Single-Shot when latency is tight (<5s), not from base-rate switching. Within each tier, one architecture is universally best	Medium-High
"SS+RAG wins some regimes"	SS+RAG was a local-corpus evidence-augmented single-shot prototype, not a standalone architecture. Its 44% win rate is an ESCALATE abstention artifact. If ESCALATE carries cost, SS+RAG wins zero regimes	High
"Evidence augmentation (RAG) does not work for verification"	2 × 3 matrix is complete with generic TF-IDF: MoA benefits (+0.170 F1), Voting benefits marginally (+0.030), SS is hurt (−0.232). Curated, source-aware evidence (FEVER/SciFact) not yet tested. N=10 per cell	Medium
"Policy sensitivity analysis shows cost_ratio dominates"	Analysis used mock-mode metrics (TF-IDF+LR), not real API metrics	Medium

Claim	Why not to overstate	Risk level
"System fits 5-second trading window"	Single-Shot does; Voting and MoA do not — paper must be clear about which architecture is being discussed for which latency budget	Low
"API non-determinism at temperature=0.0 is a concern"	Only observed on political domain; may be domain-specific rather than general	Low

Appendix: Recommended Next Experiment After Meeting

Phases 11 and 12a are complete. The 2×3 factorial matrix is fully filled (420 total API calls). Depending on professor feedback on question 2 (the A/B/C/D/E direction), prioritize one of:

- If professor prefers Latency-Aware Routing (A):** Implement Phase 12b — **RoutingPolicy** with two-tier rule (SS for $<5s$, MoA+RAG or Voting for $\geq 5s$). Compare routing against fixed baselines. Deliverable: routing performance table with expected P&L.
- If professor prefers Action Thresholding (B):** Focus on confidence-gated abstention policy using MoA+RAG (best combination). Compute optimal thresholds from cost-sensitive framework. Deliverable: PPV-operating curves with threshold recommendations.
- If professor prefers Statistical Validation (C):** Scale MoA+RAG and Voting to $N=50/\text{domain}$ ($\sim \$0.05$ total). MoA+RAG's perfect precision at $N=10$ may indicate ceiling effects. Deliverable: 95% confidence intervals.
- If professor prefers Curated Evidence (D):** Build a curated retrieval corpus (FEVER-style) and test whether it further improves MoA+RAG or rescues SS+RAG. Deliverable: before/after evidence quality comparison.
- If professor prefers Combination (E):** Recommend A+B (routing + action thresholds) as the most complete direction — they form a complete system (choose verifier $\rightarrow$ choose action) with the 2×3 matrix providing the empirical foundation.

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